Production-Scale Vacation Rental System Architecture

Resilient Cloud Architecture for Listing, High-Res Media Streaming, Real-time Pricing & Booking

Playpower Labs Take-Home Spec | Next.js 15 App Router | Edge CDN & Microservices | 99.99% Availability SLA

1 Client Layer

Desktop Web Client — APP ROUTER

Next.js 15 App Router, React 19 Server Components, Client Island hydration, Tailwind CSS v4 design system.

Next.js 15 | Tailwind v4

Interactive Photo Tour — MODAL

Category pill anchor navigation, IntersectionObserver dynamic category tracking, smooth auto-scrolling.

Category Anchors | Focus Trap

Full-Screen Lightbox — VIEWER

Body scroll lock hook, Left/Right arrow keyboard listener, Escape key dismiss, focus restoration.

KeyboardNav | ScrollLock

Booking & Pricing Widget — SIDEBAR

Sticky position booking calculator with guest selector, coupon discounts, fee breakdown.

Sticky CSS | Live Calc

2 Edge Network & CDN

Anycast Edge DNS — CLOUDFLARE

Global latency-based routing, automatic failover across 300+ Points of Presence worldwide.

Anycast | GeoDNS

Edge Security & WAF — SECURITY

Cloudflare Bot Management, L3-L7 DDoS Shield, Rate Limiting, TLS 1.3 termination, Brotli compression.

WAF | Anti-Bot

Image CDN — MEDIA

Dynamic AVIF/WebP image transformation, responsive srcset negotiation, immutable cache (max-age 1y).

Cloudinary/R2 | WebP/AVIF

3 Gateway & Services

Envoy / Kong API Gateway — ROUTING

JWT authentication, rate-limiting tokens, circuit-breaking, distributed OpenTelemetry trace propagation.

Envoy | JWT | OTel

Listing Catalog Service — MICROSERVICE

Manages property metadata, amenities, host details, house rules, and verified category taxonomy.

Go Microservice | Read Replicas

Pricing & Calendar Engine — FAST ENGINE

Real-time availability calendar, dynamic night rates, seasonal discounts, reservation hold locks.

Redis Locks | Sub-ms Query

Booking & Payments Service — TRANSACTION

Idempotent checkout flow, Stripe/Razorpay webhook orchestration, booking confirmation events.

Idempotency | Stripe API

4 Data & Storage

PostgreSQL Cluster — RELATIONAL

Primary-replica configuration, partitioned listings, bookings, user profiles, and review records.

PostgreSQL 16 | Multi-AZ

Redis Cluster — CACHE & LOCKS

Distributed cache for fast listing retrieval, booking checkout session holding (TTL 15m), rate limiting.

Redis 7 | Redlock

Elasticsearch / OpenSearch — SEARCH

Geo-spatial location indexing, full-text amenity search, fuzzy filtering by guest capacity and dates.

OpenSearch | Geo-point

AWS S3 / Cloudflare R2 — OBJECT STORE

Original high-res RAW photo assets, versioned media manifests, CDN origin storage with signed URLs.

AWS S3 | Cross-Region

5 Observability & DevOps

Datadog APM & RUM — MONITORING

End-to-end distributed tracing, Real User Monitoring (RUM), Core Web Vitals (LCP < 1.2s, CLS 0, INP < 50ms).

Datadog | Web Vitals

Prometheus & Grafana — METRICS

Cluster CPU/Memory, network I/O, database connection pooling, P99 latency alerts.

Prometheus | Alertmanager

GitHub Actions & Docker — CI / CD

Automated Playwright visual regression tests, TypeScript typechecking, zero-downtime blue/green deployments.

Playwright QA | Kubernetes